



Holography

PHYS3114

Operating Instructions

1 Safety

This holography experiment makes use of a Class 3B visible (532 nm) laser which emits levels of optical radiation that are potentially hazardous to the human eye. Specifically, the direct (intra-beam viewing) of the laser light can lead permanent eye damage or blindness. We have made every effort to manage this hazards through engineering and procedural controls, however failure to follow correct protocol can lead to increased risk of exposure. Do not attempt any experiments before you have had the appropriate safety induction and always follow the directions of your demonstrator.

- Always remove watches and jewellery from your hands and wrists when working around the open laser beams.
- Never direct the beam out of the experiment plane.
- Ensure all stray light is terminated at the bench using baffles.
- Never remain in the room when the holographic plates are being exposed.

1.1 Responsible use of equipment

- Do not remove any of the optical components directly in front of the laser. Long coherence length (single frequency) lasers can be easily damaged when laser light is directed back towards the exit aperture. An optical isolator mounted directly in front of the laser is used to prevent such occurrences.
- Care should be taken when handling the holographic plates to avoid cuts and lacerations.
- Optical surfaces and coatings can be easily damaged if touched.
- Optical mounts with actuators are precision instruments and should be handled carefully.
- Avoid leaning on the optical bench to keep the vibration isolation table level.

2 Experimental Apparatus

2.1 Optical bench

Our holography experiments are conducted on an optical table with vibration isolation legs. This is needed because the interference fringes projected onto the holographic plate needs to be stable over the duration of the exposure, which can be several minutes. The isolation legs consist of a series of pistons that are elevated by compressed air. The piston heights are regulated by pneumatic levelling valves. These adjust the intake of compressed air to compensate for changes in height. If you lean on the table, you may hear air escaping from the valves. An air compressor under the table supplies the pressurised gas for the legs. If the storage tank is depleted sufficiently (as observed by a drop in pressure) then a compressor will engage to replenish the tank, which can be loud and somewhat startling.

2.2 Laser

We use as our coherent light source a frequency-doubled diode-pumped solid state (DPSS) laser emitting at 532 nm with a maximum power of 218 mW. This laser is chosen for a couple of reasons. Firstly, the wavelength matches the sensitivity region of the holographic recording medium which means we need not use any more power than is necessary (a safety consideration). Secondly, the high powers can be used to achieve short exposure times. Why might this be helpful? Finally, the coherence length is extremely long (> 100 m), so we need not concern ourselves with issues of temporal coherence when arranging our optics.

2.3 Interlock system

The laser interlock system is used to control access to the holography room when the laser is in operation. The laser interlock system connects the laser power supply to a door-switch at the entrance to the room via a Lasernetmet ICS-6 control box. The laser interlock system must be armed (via a key and button) before the laser will operate. Under normal operating conditions the laser power is set to 2 mW, which should provide sufficient light to configure the optical arrangement. If the door is opened whilst the laser is operating, the control circuit will cause the laser to be shut off, thereby limiting the possibility of accidental exposure. The door interlock may be temporarily bypassed using the a interlock control app on the computer tablet provided. To record a hologram, you need to exit the room first (using the bypass) and then initiate the exposure sequence via the app. This will then turn the laser to maximum power (218 mW) before opening the shutter for a pre-set exposure time.

2.4 Neutral density filter wheel

Most recording configurations require the laser to be split into two separate beams with different intensities. One is used to illuminate the object / scene, whilst the second is used to provide uniform illumination across the plate. To achieve this we use a reflective graded

neutral density wheel on a rotation mount that transmits and reflects different fractions of light depending on its orientation.

2.5 Spatial filter

The spatial filter is an optical component used to achieve spatially uniform illumination over a wide area. It consists of a high magnification objective lens together with a pin-hole aperture and three-axis micro-positioning system. Under correct alignment, the pin-hole aperture is co-located at the focus of the objective and the size of the aperture is chosen to pass the focussed gaussian beam, whilst blocking higher spatial frequencies. The spatial filter is somewhat tricky to align and it is a good idea to seek assistance from your demonstrator.

2.6 Other optical components

You are also provided with a number of optical components (mirrors, lens, diffusers, etc.) that are used to establish the correct illumination conditions. The components use either a switchable magnetic base or a magnetic base that can be fixed to the table with a clamp. Any gross alignment of the optical components should be undertaken with laser shuttered. Ensure that all posts have been secured with the appropriate thumb screws before un-shuttering the beam.

2.7 Power meter

A calibrated power meter (Thorlabs, PM100D) and large area silicon photodiode (Thorlabs, S120C) is provided for you to measure the light intensity at various points along the beam path. The power meter has a maximum power range of 50 mW for 532nm CW light and so it shouldn't be used with the laser operating at full power. The active area of the photodiode is 9.5 mm in diameter.

2.8 Holographic plates

The holographic plates we use are based on a self developing polymer (LitiHolo, C-RT20 Hologram Film). This greatly simplifies the recording process that previously required multiple wet processing steps (developing, bleaching, fixing, etc.). Failure of the developing process was a common source of frustration for budding holographers and frequently led to disappointment. Another nice feature of these new types of holographic plates is that they have a reasonably high threshold for photo-polymerisation. This means that setting up the plates can be done in a reasonably well lit room without fear of ruining the plates. The following specifications are provided by the manufacturer (<http://litholo.com/hologram-film.html>):

Specification Sheet (C-RT20)	
Specifications	Substrate: 2.0mm glass Film thickness: 16 microns Coversheet: 60 microns, TAC
Developing Process	Self-developing holographic photopolymer develops as it exposes. After full exposure, material is saturated and no longer sensitive to light. An optional post-processing exposure to a strong spot light or with film plate held near a bare light bulb may increase hologram brightness and further bleach any residual colour.
Performance	DE Transmission: >90% holographic grating test DE Reflection: >90% holographic mirror test Sensitivity: 400 - 690 nm Exposure Energy (635nm): 20 mJ/cm ² Exposure Energy (532nm): 30 mJ/cm ² Exposure Energy (450nm): 80 mJ/cm ²
Storage	Temperature: 55 - 80 degrees F (do not refrigerate) Humidity: <50% RH Lifetime: 12 month warranty minimum

3 Experiment 1: Denisyuk Hologram

A Denisyuk hologram is a recording configuration where the reference beam also provides the light for the object illumination. It is a type of reflection or volume hologram that is suitable for small objects mounted close to the plate. It will allow you to focus on correctly establishing the reference beam and setting up the holographic plate. Also, it almost always works, which is great for morale.

The optical arrangement for recording the Denisyuk hologram is provided in Fig. 1. The laser is mounted at the back of the table pointed away from the door. An optical isolator is mounted approximately 15 cm from the exit aperture followed by the mechanical shutter (interfaced with the computer tablet). The space between the laser aperture and the optical isolator is enclosed by a laser tube to protect against accidental exposure. From this point you should now be able to configure your experiment and the following steps are provided as a guide. Some of the steps may already be done for you, but it is good to confirm that the configuration is correct.

3.1 Experimental Procedure

1. Use the power meter directly after the shutter to verify the output power from the laser is set at 2 mW.
2. Mount the mirror at the end of the table and reflect the beam by 90 degree. It is

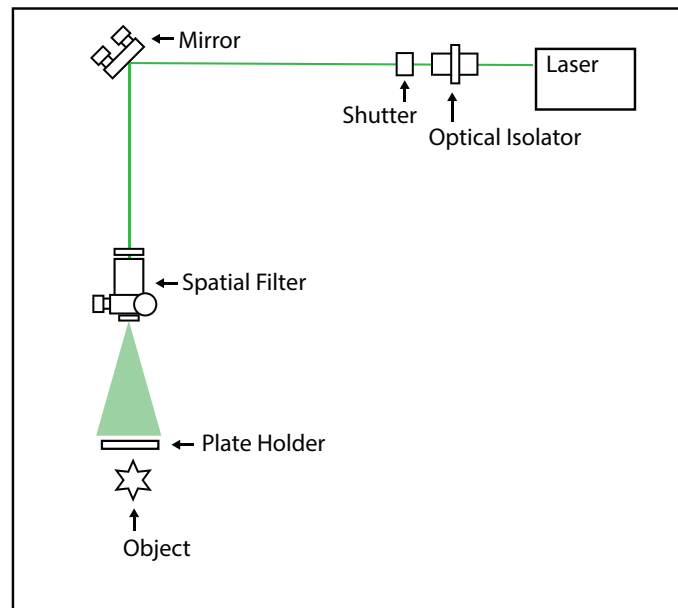


Figure 1: The optical configuration for the Denisyuk hologram configuration. The laser beam exiting the shutter is directed towards a spatial filter using a mirror on a kinematic mount. The spatial filter produces a uniform illumination across the plate. The object mounted behind the plate reflects light back towards the plate creating an interference pattern that is recorded within the volume of the film

helpful to use the holes on the optical table as a guide.

3. With the pin hole removed, mount the spatial filter on the table and align the optical axis to be centred and parallel with the laser beam path. It is helpful to use the light reflected from the objective lens inside the filter to orient the spatial filter (as the demonstrator for assistance). Once complete the beam coming out of the filter should appear as an expanding cone of light centred around the same beam path as the incoming light.
4. Insert the pin hole aperture and, by monitoring the transmitted light pattern, adjust the aperture position to achieve a uniform output. This is a non-trivial step for the untrained user, so please seek assistance.
5. With the spatial filter aligned place the film holder at a distance and height where the mounted plate will be fully illuminated.
6. Dim the lights and use the power meter at the optical plate to estimate the intensity of the reference beam at the plate. Estimate the exposure time required for recording, recalling that the laser will be 108 times brighter during the exposure. Due to ageing of the holographic plates, we recommend doubling the calculated exposure time.
7. Place the object stand directly after the plate holder and position your object so it will be centred with respect to the plate. You can use a pre-exposed plate to help position your object correctly.

8. Place light baffles at positions on the bench to obscure any bright spots that may otherwise appear on the recording.
9. At this point you should turn down the lights, shutter the beam, and obtain an unexposed holographic plate from the laboratory staff or demonstrator. Discuss your exposure plans with our demonstrator
10. Place the with the plate with the photosensitive film facing the object in the plate holder.
11. Use the app to override the door interlock and exit the room.
12. Set the exposure time and initiate the recording sequence. This consists of a delay for 3 minutes that allows the airflow in the room to stabilise and the laser to reach a stable maximum, after which the shutter is opened for the set time. Once the exposure is completed the laser will turn back down to low power mode.

3.2 Analysis Questions

At this point you should have recorded your first hologram. Wow, congratulations! The hologram should be viewed in reflection (i.e. with the light behind you), using a point like white light source such as a halogen lamp. In the absence of sunlight, you can use the torch from a mobile phone. An additional optional post processing step is to leave the plate under a bright light (e.g. direct sunlight) for approximately 10 minutes to improve the brightness of the hologram. When the bluish tone disappears,

Whilst you are enjoying your wonderful creation, you should answer the consider the following questions when discussing your results:

1. How is the information about the scene / object captured by the plate?
2. Why is this considered a volume hologram?
3. What are the coherent requirements for the light source in recording this type of hologram (remember that coherence is not an all or nothing parameter)?
4. Why is it that you can use an incoherent light source to read out the hologram?
5. Why is it that the hologram changes colour when the viewing angle is modified?
6. Explain the basic properties and physical origin of the pseudoscopic image, which is observed when the hologram is viewed from the reverse side of the plate.

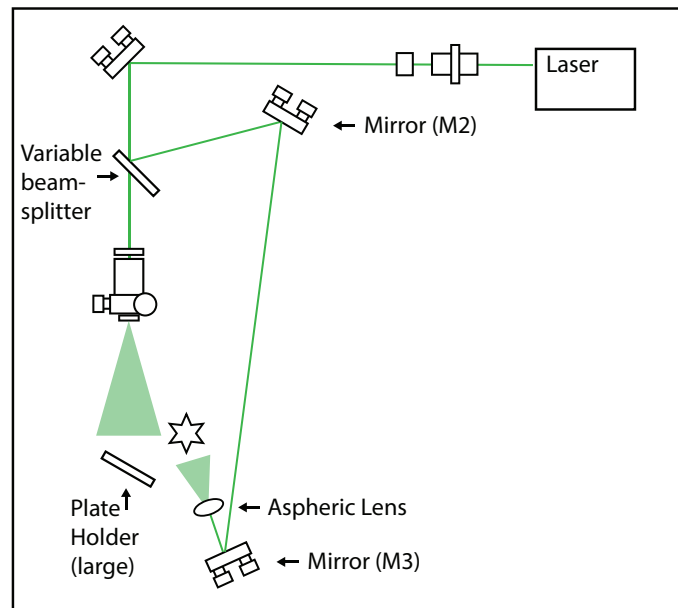


Figure 2: The optical configuration for the Transmission hologram. A variable beam splitter placed before the spatial filter is used to create a second beam for illuminating the object. The object is also colocated to same side of the plate as the incoming reference beam. A short focal length aspheric lens is used to expand the object beam to fully illuminate the scene.

4 Experiment 2: Transmission Hologram

In this next experiment we will modify the experimental arrangement to include a separate object beam and locate the object such that the reference and object beams illuminate the plate from the same side. We will use the large holographic plate for this experiment so that two different exposure conditions can be done on the one plate. A schematic of the optical layout is provided in Figure 2

The following steps may be used to establish the correct arrangement. It assumes you already have the Denisjuk configuration set. If this isn't the case, you may need to review the steps in the previous experiment.

4.1 Experimental Procedure

1. Replace the single plate holder with the large holographic plate holder. This holder is designed to allow two separate exposures on the one plate. The demonstrator will show you how it works.
2. Tilt the plate holder such that the reference beam is incident at an angle of about 45° from the normal. The expanded reference beam should still illuminate the entire plate.
3. Reposition the object stand such that it is in front of the holographic plate. The aim is get the object centred with respect to the plate, but at distance where the reference beam won't intersect with the object.

4. With the beam shuttered, insert the neutral density filter wheel at an angle of approximately 45° and place a baffle to block the reflected beam.
5. With the beam un-shuttered, adjust the orientation of the filter wheel so that the deflected beam is incident on the mirror (M2).
6. Using the power meter to monitor the deflected beam, adjust the filter orientation of the filter such that the relative power between the reference and object paths is 4:1. Note that the filter wheel has a small precession associated with the rotation that can lead to misalignment, so ensure the beam does not walk off the power meter.
7. Once the filter wheel is in position, there will be a slight misalignment of the spatial filter (why?). This should be easily corrected by adjusting the in plane actuators.
8. Adjust the orientation of M2 to direct the beam towards the M3 mirror, and then adjust the orientation of M3 so that the beam is incident on the object.
9. Insert the aspheric lens into the object beam path to expand the beam so that it fully illuminates the object. Adjusting the separation between the aspheric and object can be used to optimise the illumination.
10. Once you are happy with the arrangements, dim the room lights and use the power meter to measure the intensity at the holographic plate and estimate the exposure time for your configuration.
11. Shutter the beam, and obtain a large holographic plate. Discuss your exposure plans with the demonstrator before you begin.
12. Place the with the plate with the photosensitive film facing the object in the plate holder.
13. Use the app to override the door interlock and exit the room.
14. Set the exposure time and initiate the recording sequence. This consists of a delay for 3 minutes that allows the airflow in the room to stabilise and the laser to reach a stable maximum, after which the shutter is opened for the set time. Once the exposure is completed the laser will turn back down to low power mode.
15. Once complete, you should use the override to re-enter the room and adjust the plate position and scene (if required) for the second exposure.

4.2 Analysis Questions

Congratulations, you have now made your first transmission hologram. To verify that the hologram has worked, you can remove the object and view the hologram with the reference beam. The reference beam should produce a diffracted field that reconstructs the scene and that can be viewed from different angles. The scene can also be viewed using a narrow spectral light sources such as a coloured LED. A light source with three different coloured LEDs is provided for you to illuminate your hologram and observe the influence of different viewing conditions.

The consider the following questions when discussing your results:

1. How are the fringe recordings differ between this transmission hologram and reflection hologram recorded in the previous experiment.
2. Why can a transmission hologram not be observed using white light illumination?
3. Explain the difference between a phase and amplitude hologram? Which is relevant to your experiments.
4. What are the coherence requirements of the light source for recording and reading the hologram.
5. Provide a physical interpretation of observed changes to the reconstructed image when the wavelength of the illumination is changed.
6. What physical parameters define the spatial resolution of the hologram.
7. Observe the effect of illuminating just a small section of the hologram. This can be done by illuminating through a mask with a small window or variable aperture. How would you interpret these observations?

5 Experiment 3: Double-exposure holographic interferometry

Using the transmission hologram configuration we can next explore how holograms can be used to perform precise interferometry on a 3D object. The procedure is to make a recording of an object in two stages: the first half of the exposure is done to capture the reference conditions and the second exposure records the same object under some perturbation. We will use an aluminium can that has been painted white as our object, and the perturbation will take the form of a weight applied to the top that will provide stress induced deformation. In practice, we will first expose the stressed object and then remove the weight.

1. Remove your transmission object and replace it with the white-painted aluminium can. The can is fixed to a heavy metal base that can support the additional weights used to stress the can.
2. Adjust the position to optimise the illumination conditions and its visibility through the holographic film holder. You may need to reposition the object beam mirror (M3) and aspheric lens.
3. Place a weight (approx. 750 g) on top of the can.
4. Dim the lights and use the power meter to calculate the intensity of the plate and determine the required exposure time (remembering to apply the appropriate scaling factors).
5. With everything in place, shutter the beam, dim the lights, and insert the holographic plate.

6. Exit the room using the interlock override and, with **half** the exposure time set, initiate the exposure sequence.
7. Enter the room and carefully remove the weight from the top of the can, making sure not to disturb anything else.
8. Repeat the exposure sequence with the other half of the exposure time set.
9. Replace the weight on the can and repeat the double exposure hologram. Try using a different exposure time or a different ratio of times for the stressed and unstressed objects.

5.1 Analysis Questions

If the double exposure hologram worked, you should observe the can with a series of interference fringes superimposed on the object. Can you explain this observation. Can you provide any quantitative analysis of the effect of the weight on the can?

6 Optional Experiment 4: Reflection hologram

In this experiment you will create a second reflection hologram, but using a separate reference and object beam. The optical set-up is depicted in Figure 3. The arrangement is similar to that of the transmission, but with the object located behind the plate. The instructions otherwise follow similar to the transmission hologram set-up, but the ratio of reference to object beam should be changed to 2:1.

7 Experiment 5: Michelson Interferometer

The configuration for the transmission and reflection holograms is akin to a very large Michelson interferometer. Because the laser has a very long coherence length, we need not concern ourselves with the relative distance of each of the interferometer arms, however other factors can play a role in recording interference fringes (e.g. environmental conditions). For this experiment, you will use the components of holography set-up to construct an standard interferometer with comparable dimensions to recording geometry. You can observe the resulting interference fringes projected onto a viewing screen, check the fringe contrast and also see what happens when the system is perturbed. The suggested experimental arrangement is shown in Figure 4.

7.1 experimental procedure

1. Start by removing the unnecessary components from the bench.
2. Adjust the position of mirror M1 so that it is approx. 50 cm from the edge of the bench.

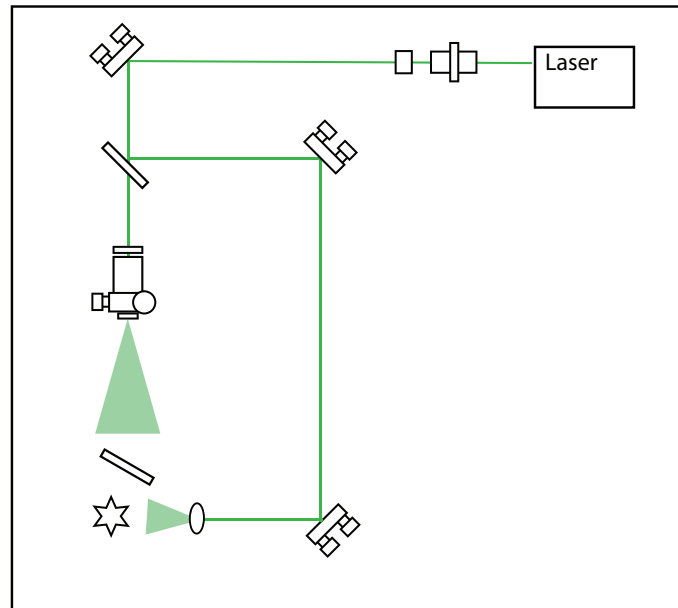


Figure 3: The optical configuration for the Reflection hologram. A variable beam splitter placed before the spatial filter is used to create an a second beam for illuminating the object. The object is place behind the plate and a short focal length aspheric lens is used to expand the object beam to fully illuminate the scene.

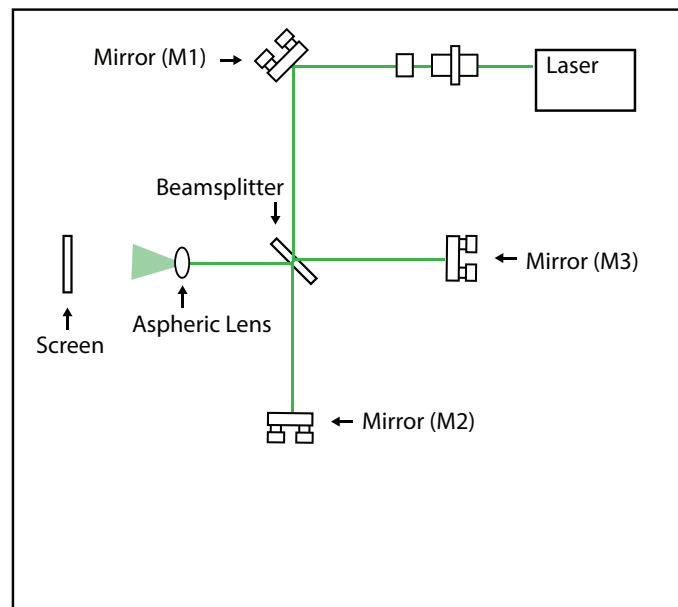


Figure 4: The optical configuration for Michelson interferometer. The incoming beam is split into two using a plate beam splitter mounted such that the angle of incidence of the beam is at 45° to the normal. The two beams are allowed to propagate for some distance before being reflected back towards the on beamsplitter with mirrors. The transmitted/reflected beams are then projected onto a screen to observe the interference pattern.

3. Orient the mirror so that reflected beam is travelling parallel to a row of mounting holes on the bench.

4. Place the mirror M2 at the end of the bench orient so that the beam is at normal incidence.
5. Place the plate beamsplitter in at about the midpoint between the first two mirrors and orient it so that the reflected beam is perpendicular to the incident beam.
6. Position the mirror M3 at about the same distance from the beamsplitter as M2 and orient the mirror so that the beam is at normal incidence.
7. The beams originating from mirrors M2 and M3 should now be passing back towards the beamsplitter and some fraction of each beam will be directed towards the viewing screen and propagating parallel.
8. Place the aspheric lens in line with the beams to expand the beams onto the viewing screen.
9. Make small adjustments to the beam paths to maximise the observed fringe separations.
10. Observe the behaviour of the fringes when the set-up is disturbed by air flow, vibrations and stimuli.